

Multimodal Knowledge-infused VLM for Respiratory Disease Prediction and Clinical Report Generation

Mukhlis Raza, Saied Salem, Hyunwook Kwon, Jamil Hussain, Yeong Hyeon Gu, Mugahed A. Al-antari

Abstract—Accurate and interpretable computer-aided diagnosis (CAD) for respiratory diseases remains a significant challenge due to the complex multimodal nature of medical data. Existing systems often fall short by focusing on unimodal retrieval analysis or lack sophisticated retrieval strategies, which limit both diagnostic reliability and interpretability. To address this, we propose a novel end-to-end CAD framework that employs a large language model (LLM) with advanced retrieval-augmented generation (RAG) techniques, leveraging multimodal retrieval and structured radiology report generation. The framework employs a dedicated vision encoder for binary identifying view types (frontal anterior-posterior (AP), Lateral View (LV)) and multi-label classification for 14 disease categories, with Grad-CAM providing visual interpretability. Concurrently, an advanced text encoder processes unstructured medical reports, enabling the generation of structured zero-shot reports. We propose three novel RAG approaches to enhance retrieval accuracy and reduce LLM hallucination by improving contextual alignment for medical report generation: (1) Caption-Augmented Fusion (CAF), (2) CLIP-Aligned Fusion (CLAF), and (3) Domain-Grounded Fusion (DGF). The performance of our radiology report generation (RRG) is rigorously evaluated using a comprehensive suite of lexical, semantic, and clinical metrics, including METEOR, ROUGE-L, BERTScore, and CheXpert Labeller. The clinical relevance and diagnostic accuracy are further validated by a panel of expert radiologists, establishing a gold standard. Our proposed CAD system can accurately diagnose respiratory conditions and generate structured radiology reports within 22 seconds. This rapid and accurate system holds considerable promise for real-world clinical applications and is well-suited to meet the demands of modern medical practice. The GitHub code is available here: <https://github.com/AISSLab2025/MedXRay-CAD>

Index Terms— Retrieval-Augmented Generation (RAG); Multimodal Retrieval; Vision-language Model (VLM); Vision Large Language Model (VLLM); Computer-aided Diagnosis (CAD); Radiology Report Generation (RRG).

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I. INTRODUCTION

LARGE language models (LLMs) have transformed natural language processing with strong performance in understanding complex medical concepts and generating human-like responses [1]. However, in clinical diagnostics, especially radiology, LLMs face major limitations due to hallucinations the generation of plausible but incorrect information which pose serious patient safety risks [2]. This is critical in interpreting chest X-rays for diagnosing respiratory diseases affecting over 545 million people globally as of 2017 [3]. LLMs are based on fixed training data, which leads to outdated knowledge and a lack of access to the latest medical evidence. The retrieval-augmented generation (RAG) contributes to this through the incorporation of external knowledge bases to minimize hallucinations and improve the accuracy of diagnosis [4]. However, RAG has been used textual or unimodal retrieval and does not reflect the complexity of the visual-textual relationships present in medical imaging. To overcome these shortcomings, multimodal RAG has been proposed in generalizing traditional text-based RAG with visual and text insights. This is particularly useful in radiology, where diagnostic decisions are based on a synthesis of image findings, textual reports, patient history, and clinical guidelines [5]. Recent work demonstrates that complementary modalities of multimodal RAG perform better than unimodal approaches in medical questions and answering tasks [6]. However, the existing systems have some challenges, such as weak cross-modal matchings, incoherent retrieval pipelines, and the risks of hallucination in synthetic scenarios. Such problems may lead to an incomplete diagnosis and unsafe clinical recommendations [7, 8]. Overcoming them requires advanced fusion strategies to align visual diagnostics with textual knowledge while ensuring factual grounding. To address these gaps, this study proposes a novel retrieval-

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fuse-generate (RFG) framework that integrates advanced vision encoders with three specialized multimodal RAG approaches to comprehensively evaluate and demonstrate the impact of different forms of multimodal retrieval on clinical report generation. By systematically testing text-based (Caption-Augmented Fusion (CAF)), text-image fusion (CLIP-Aligned Fusion (CLAF)), and domain knowledge adoption guided retrieval (Domain-Grounded Fusion (DGF)) methods, we can rigorously compare how various combinations of textual and visual information contribute to the accuracy, contextual relevance, and factual grounding of the generated reports. This approach not only validates the versatility of our multimodal framework but also crucially establishes that effective radiology report generation requires leveraging both textual and imaging data, rather than relying on text alone. The key contributions are outlined here:

1. A novel computer-aided diagnosis (CAD) system is proposed by integrating image and text interpretation to accurately predict multi-label diseases from chest X-ray (CXR) images-texts and generate structured radiological reports leveraging the power of VLM, VLLM, and RAG.
2. Three novel advanced multi-modal RAG strategies (CAF, CLAF, and DGF), concatenating with multi-label classification methods to bridge image and text diagnostic features, optimizing multi-modal information retrieval.
3. A multi-vision encoder enhances XAI models with saliency maps for better CAD outcomes, improving radiological report generation by providing accurate disease predictions, refining RAG retrieval, and guiding VLLMs to generate relevant text explanations.

II. RELATED WORK

A. Vision Encoders for Chest X-ray Classification

Vision encoders are very important for computer-aided diagnosis systems for respiratory disorders. Model like DenseNet121 and ResNet50, with their step-by-step feature extraction technique, along with dealing with feature loss, have gained popularity as deep convolutional neural networks in automated multi-labelled classification of chest X-rays [9, 10]. But a recent study has emerged outlining the shortcomings of individual deep learning models in medical imaging. Haque et al. [11] demonstrate differences in the performance of DenseNet121 depending on the resolution of the images as opposed to differences in fairness in Seyyed-Kalantari et al. [12] across demographic groups. Such problems indicate biases in data and dependence on ground-quality inputs. Besides, most approaches only apply single-view analysis and disregard compliant multi-view data (such as frontal and lateral chest X-rays) and focus mainly on classification accuracy, which does not utilize visual embedding and retrieval to enhance knowledge-based diagnosis.

B. RAG in Medical Applications

RAG has been increasingly used on medical tasks to reduce the prevalence of LLM hallucinations by grounding generated output in external knowledge. Ranjit et al. [13] introduced CXR-RAG, which retrieves radiology text using a contrastively

pretrained vision-language model and generates reports with GPT-3.5-Turbo. While this approach minimizes hallucinations, its reliance on unimodal (text-only) retrieval is a key limitation, as it can introduce noisy or irrelevant information when the textual context alone is insufficient to capture the full diagnostic picture. Similarly, Huang et al. [14] proposed RagPULSE, a Distill-Retrieve-Read framework for medication consultation that summarizes dialogues into search queries to improve evidence retrieval. However, it is also limited to retrieving textual information and does not generalize to multimodal medical data, making it unsuitable for complex radiological scenarios that depend on visual evidence. In contrast, our framework overcomes these limitations by employing true multimodal retrieval, which provides a more complete and contextually grounded basis for report generation.

C. Multimodal Report Generation

Multimodal approaches for radiology report generation integrate visual and textual data to improve clinical accuracy by incorporating auxiliary information or relational knowledge to enhance report quality. For instance, Zhang et al. [15] propose a joint probability graph reasoning method that leverages disease co-occurrence patterns via a static probability matrix learned during training. Their model enhances visual feature reasoning and cross-modal fusion through a gating mechanism. However, it relies on fixed relational knowledge rather than dynamic retrieval at inference, making it a feature-enhanced generative model, not a retrieval-augmented one. Similarly, Wang et al. [16] proposed R2GenCSR, a framework that enhances image features by retrieving context samples from the training set. This method acts as feature-level data augmentation during training, not as a dynamic RAG approach. Moreover, Xia et al. [17] developed MMed-RAG, a system aimed at improving the factuality of Medical Large Vision-Language Models. While advanced, its performance depends heavily on the quality of its initial fine-tuning, and it retrieves modalities in separate streams before fusion, which can lead to fragmented and misaligned context. Lastly, Castro et al. [18] introduced PadChest-GR, a high-quality curated dataset with grounded findings for training models. It is a dataset, not a generation method, and does not offer a dynamic retrieval strategy. Our approach, therefore, stands apart by performing dynamic, case-based multimodal retrieval at inference time, ensuring that the generation process is guided by the most relevant and complete prior clinical cases.

III. METHODOLOGY

The proposed end-to-end retrieval-fuse-generate (RFG) hybrid CAD framework is designed to better integrate visual and textual embeddings for generating radiology reports, which is a key missing feature in the existing systems. existing systems. The system works by using specialized encoders to create multimodal embeddings from images and text. We then use these embeddings and our proposed retrieval techniques to find the most relevant clinical knowledge from our vector database. This ensures the final reports are both clinically accurate and well-grounded in the given context. As shown in Fig 1., the proposed end-to-end multimodal CAD framework is

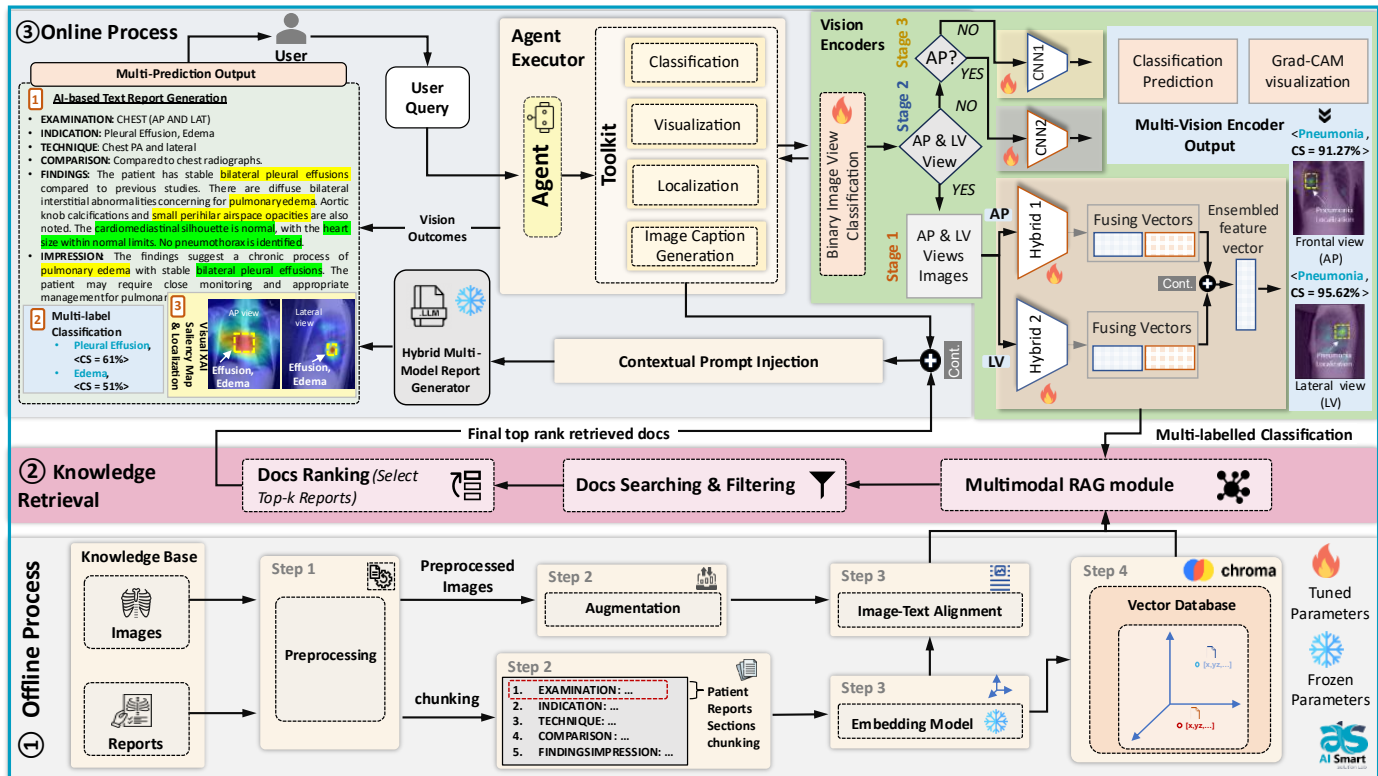


Fig 1. The proposed end-to-end AI CAD framework for radiology report generation: (1) Offline Process: develops a knowledge base by preprocessing chest X-ray images and reports into multimodal embeddings; (2) Document Retrieval: acquires relevant information using knowledge-infused RAG techniques; and (3) Online Process: Real-time diagnosis and report generation. The CAD system uses an agent with specialized tools, including vision encoder (classification and XAI-Grad-CAM for visualization) and text encoder for XAI-report synthesis. The vision encoders utilize CNN1 (ResNet50), CNN2 (DenseNet121), and hybrid models that ensembled both CNNs for better performance

composed of three primary sections. First, an *Offline Process* is dedicated to creating a comprehensive knowledge base. The second component is *Knowledge Retrieval*, where our new RFG methods are used. Finally, the *Online Process* utilizes an agent and multi-view vision encoders to generate and analyze the reports. Each of these sections will be examined in detail in the subsequent parts.

A. Offline Process

1. Dataset

In this study, three open-source chest X-ray datasets are used for constructing the proposed CAD framework: (1) The **MIMIC-CXR** dataset [19] is a large, publicly available collection containing 377,110 chest radiographs with associated radiological reports from 227,000 imaging studies. Notably, while the dataset includes both AP and LV views, a subset of cases lacks LV images. It features a multi-label structure, reflecting clinical reality where patients often present with multiple concurrent pathologies. Labels for 14 common chest X-ray findings were extracted using the CheXpert labeler [20], categorized as positive, negative, uncertain, or missing. This study adopts the U-zeros labeling technique [20]. (2) The **IU-Xray** dataset [21], is also multi-labeled dataset and reflects complex and real-life cases. It includes 8,121 images paired with 3,087 brief text reports sourced from the Indiana Network for Patient Care. Each report contains findings detailing pathological conditions, impressions, and provides clinical

interpretations. (3) The **ChesX-ray8** dataset [22] comprises 108,948 frontal chest X-ray images from over 32,000 patients and includes eight diseases (single label schema per patient), but lacks radiological text reports. Importantly, each dataset follows its own expert labeling schemas making it challenge to follow the same evaluation scenario at once. Thus, MIMIC-CXR is mainly used for training, validation, and evaluation purposes, while IU-Xray and ChesX-ray8 are used to conduct the ablation studies.

2. Preprocessing

As shown in Fig 1., preprocessing is employed for enhancing the dataset during the offline process. A subset of 5,000 images is used to determine the ideal parameters for CLAHE, which is used to enhance all chest X-ray images. To increase model robustness, the images are then subjected to data augmentation, which includes center cropping, rotations up to $\pm 15^\circ$, random horizontal flips, and normalization. To identify the optimal vision encoder, an automated MLOps workflow (Weights & Biases) is employed for benchmarking eight candidate models on the subset dataset. Key hyperparameters (models, batch size, epoch learning rate, learning factor, optimizer, augmentation) are fine-tuned based on validation accuracy, precision, and loss. DenseNet121 performed well for binary image-view classification, while DenseNet121 and ResNet50 [23, 24] are chosen for multi-label disease classification. For the associated text, patients without active clinical findings (1, -1, or 0) are

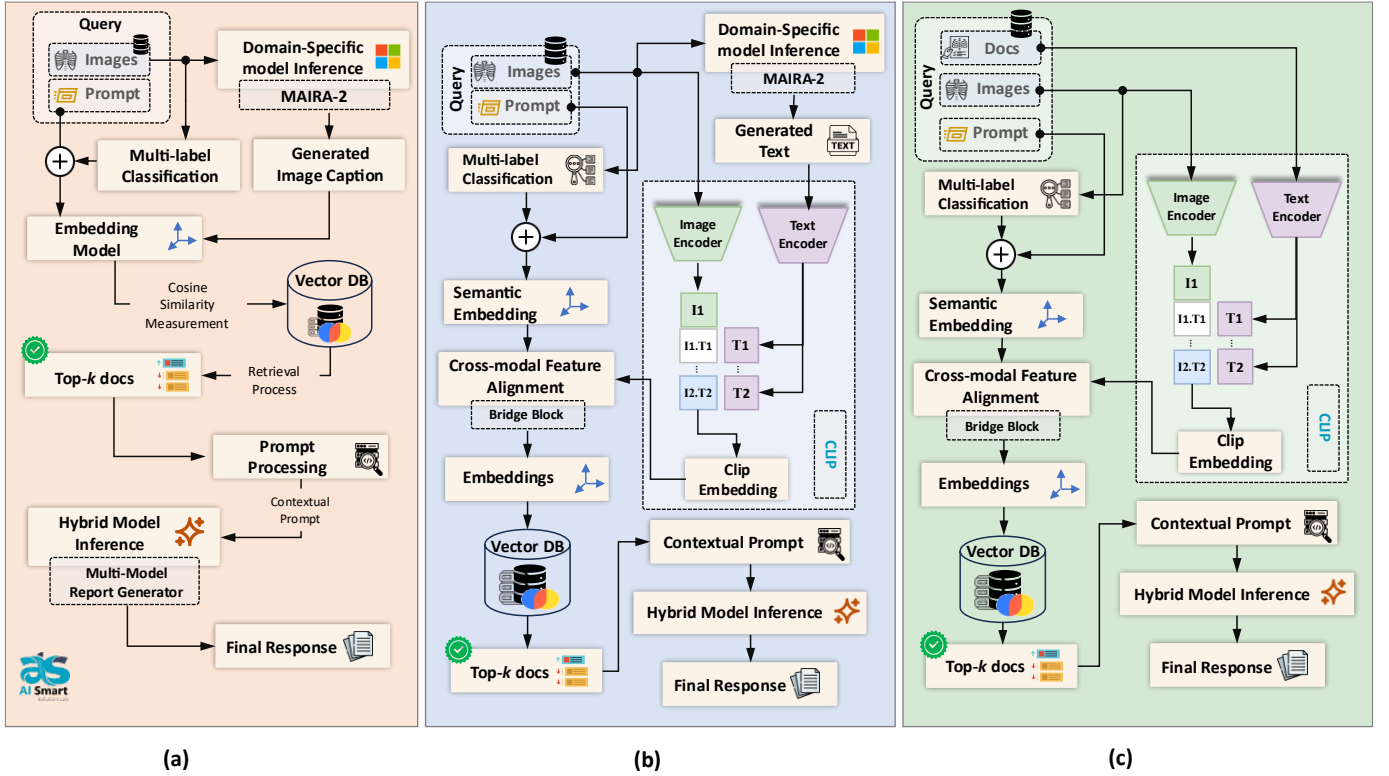


Fig 2. Knowledge retrieval overview of multimodal RAG approaches: (a) Caption-Augmented Fusion (CAF), (b) CLIP-Aligned Fusion (CLAF), and (c) Domain-Grounded Fusion (DGF). These methods combine textual and visual modalities using advanced embeddings and dimension alignment to optimize retrieval accuracy and enhance zero-shot radiology report generation.

excluded, reports section headers are standardized, key sections such as findings and impressions are extracted, and irrelevant content like doctor footnotes is removed. The curated image-text pairs are embedded to capture all relevant features and are stored in a Chroma vector database for efficient retrieval.

3. Vector Database Embedding

During the offline stage, a knowledge base is created by embedding MIMIC-CXR data set reports into a vector database to support efficient multimodal retrieval in the online stage. This process aligns visual and textual data for subsequent classification and report generation tasks. As shown in **Fig 1.** (Offline process), the textual data from radiology reports is preprocessed by chunking into six standardized sections $S \in \text{sections}$ include Examination, Indication, Technique, Comparison, Findings, and Impression. leveraging regular expressions to identify and extract section headers and content. For each section S , the text content T_s is embedded into a d -dimensional vector using the *mxbai-embed-large* [25] semantic embedding model, which incorporates angle optimization in complex space to enhance semantic alignment and mitigate cosine saturation issues, as introduced in Li, X. et al. (2023) [25]. The embedding is computed as:

$$X_s = E_{\text{txt}}(T_s; \theta_{\text{txt}}), X_s \in \mathbb{R}^d, \quad (1)$$

Where X_s is the embedding vector, E_{txt} is the text encoder with parameters θ_{txt} , and $d = 2024$ is the dimensionality of the embedding space. Embeddings are stored in a Chroma vector database with metadata including section type and a unique case ID for traceability, leveraging cosine similarity for efficient similarity searches. Retrieval efficiency relies on cosine similarity.

$$\cos(X_i, X_j) = \frac{X_i \cdot X_j}{\|X_i\| \cdot \|X_j\|}. \quad (2)$$

The embedding model is optimized using a contrastive loss with an angular difference objective to improve subtle semantic distinctions.

$$\mathcal{L}_{\text{loss}} = \log \left(1 + \sum_{S(X_m, X_n) > S(X_i, X_j)} e^{\frac{\cos(X_m, X_n) - \cos(X_i, X_j)}{\tau}} \right), \quad (3)$$

Where $S(X_i, X_j)$ represents the reference text semantic similarity score between embeddings X_i and X_j , the summation iterates over pairs where a "negative" pair (X_m, X_n) has higher similarity than the "positive" pair (X_i, X_j) ; $\cos(\cdot)$ is the cosine similarity, and τ is a temperature parameter. The loss yields similar pairs closer and dissimilar ones further apart, increasing multimodal retrieval discriminative power. It is processed in batches of 10,000 documents to control computational burden. The retrieval returns the top-k semantically related documents by using cosine similarity and combines them in contextual prompts to the LLM. Such section-wise embedding supports fine-grained and domain-specific information retrieval, where sections in the report of interest (e.g. FINDINGS) should be prioritized by default but adaptable to ablation or query relevance.

B. Knowledge Retrieval

Multimodal retrieval improves contextual alignment between the input query and vectorized documents by leveraging both image and text modalities. As shown in **Fig 1.**, the knowledge retrieval stage is the key to our framework. **Fig 2.** details the proposed multimodal RFG strategies. These methods improve

Algorithm 1 Multimodal Medical Report Generation with Advanced RAG

Input: $Patient_{AP} = \{i_{f1}, i_{f2}, \dots, i_{fn}\}$, $Patient_{LV} = \{i_{l1}, i_{l2}, \dots, i_{ln}\}$
 // chest X-ray images $\mathcal{R}$ // Ground truth reports, C //
 generated image captions by MAIRA-2, q // Retrieval
 query, ρ // Contextual prompt for final report generation

Result: Generated Medical Reports $\hat{\mathcal{R}}$

```

foreach  $i$  in  $Patient_{image}$  do
   $cls \leftarrow$  Getting top classification for each  $i$ 
   $q = \rho \oplus cls$ 
  if  $i$  in  $C$ 
    if method =  $CAF$ 
       $C \leftarrow$  Corresponding caption  $i$ 
       $e \leftarrow$  Fused embedding of  $E_{txt}(q)$  and  $E_{txt}(C)$ 
    else // method =  $CLAF$ 
       $C \leftarrow$  Corresponding caption  $i$ 
       $e \leftarrow$  Fused embedding of  $E_{clip-text}(q)$ ,  

       $E_{clip-img}(I)$  and  $E_{txt}(C)$ 
    end
  else //  $\mathcal{R}$  for method =  $DGF$ 
     $e \leftarrow$  Fused embedding of  $E_{clip-text}(q)$ ,  $E_{clip-img}(I)$   

    and  $E_{txt}(\mathcal{R})$ 
  end
   $R_d \leftarrow$  Retrieve top- $k$  similar documents from vector  

  database using  $e$  and cosine similarity
   $q = \rho \oplus cls \oplus R_d$ 
   $LLM \leftarrow q$ 
   $\hat{\mathcal{R}} \leftarrow LLM$ 
end
  
```

retrieval accuracy in a zero-shot setting, forming a robust basis for report generation. We also evaluate base retrieval methods, RAG-Fusion and contextual compression [26, 27] to validate the effectiveness of multimodal retrieval. RAG-Fusion uses LLMs to generate questions for improved fact-checking, while contextual compression reduces document size while preserving semantic relevance, as the overall bootstrapping process is described in Algorithm 1.

1. Caption-Augmented Fusion (CAF)

This strategy fuses textual queries with captions generated from input chest X-ray images to enrich retrieval. Captions are produced using a pre-trained domain-specific model [28] encapsulating visual semantics absent in the query. Both query and caption are embedded independently using the embedded model, and their embeddings are averaged to form a composite representation for database querying. This approach dynamically incorporates visual context, improving relevance for queries involving subtle radiographic features.

$$q = \rho \oplus cls, \quad (4)$$

$$e_{CAF} = \frac{1}{2} (E_{txt}(q) + E_{txt}(C)), \quad (5)$$

Where ρ is the base prompt, cls is the multi-label classification string, $\oplus$ denotes textual concatenation, q is the fused query, C is the generated image caption by MAIRA-2 [28], and E_{txt} denotes the text encoder. The fused embedding e_{DMFGIC} is used to retrieve semantically aligned documents via cosine similarity.

2. CLIP-Aligned Fusion (CLAF)

CLAF leverages CLIP [29] for cross-modal alignment, encoding the textual query and input image alongside the generated caption. CLIP encoders produce 512-dimensional vectors for the query and image, which are zero-padded to match the 1024-dimensional semantic embedding. The three embeddings are averaged to create a unified multimodal representation, capturing intricate text-image relationships. This is particularly effective for identifying condition-specific patterns, as CLIP's vision-language pre-training facilitates precise retrieval when fused with caption semantics. However, when used for text-based database queries, CLIP's embeddings are complemented by Ollama to prioritize textual semantic quality over pure image-text pairing.

$$e_{CLAF} = \frac{1}{3} \left(\begin{matrix} \Delta(E_{clip-text}(q), d) + \\ \Delta(E_{clip-img}(I), d) + \\ E_{txt}(C) \end{matrix} \right), \quad (6)$$

Where I is the input image, $E_{clip-text}$ and $E_{clip-img}$ are CLIP's text and image encoders, respectively, $d = 1024$ is the target dimension, and $\Delta(\cdot, d)$ zero-pads to cross-modal feature alignment. The fused embedding e_{CEDMF} queries the database for top- k documents.

3. Domain-Grounded Fusion (DGF)

DGF integrates textual queries, medical images, and additional domain knowledge (e.g., structured reports or clinical guidelines) into a unified embedding to enhance factual accuracy and retrieval precision. It is an extended version of CLAF, these variants substitute generated captions with reference text annotations extracted from original reports, enhancing factual alignment and retrieval precision. Reference text is parsed via regular expressions from key sections (e.g., "FINDINGS" or "IMPRESSION"), providing clinically accurate descriptions. Embeddings for the query, image, and reference text are fused similarly, mitigating errors from caption generation models. This method excels in annotated datasets, ensuring that retrieved documents closely match verified radiological insights.

$$e_{DGF} = \frac{1}{3} \left(\begin{matrix} \Delta(E_{clip-text}(q), d) + \\ \Delta(E_{clip-img}(I), d) + \\ E_{txt}(\mathcal{R}) \end{matrix} \right), \quad (7)$$

where $\mathcal{R}$ is the reference text. The fused embedding $e_{CEDMF-GT}$ enables retrieval of highly relevant documents, reducing hallucination risks in report generation.

C. Online Process

As shown in Fig. 1, the online part of the proposed report generation system uses a smart agent to dynamically invoke a vision encoder and to compose real-time radiology reports. Through this process, multi-label classifying results are incorporated with multimodal retrieval, which leads to contextually rich and clinically accurate reports. This section describes the multi-vision encoder and the radiology report generation (RRG) pipeline; this covers real-time query computation and synthesis of reports.

1. Multi-vision Encoder

A DenseNet121 model is initially employed to classify images by view (frontal or lateral), with the dataset divided into 70% for training, 20% for validation, and 10% for testing. Three distinct classification stages are implemented to accommodate

varying image availability: (Stage 1) both AP and LV views available, (Stage 2) AP-only, and (Stage 3) LV-only. For Stage 1, two parallel pipelines process AP and LV images using DenseNet121 and ResNet50, respectively, with their output features ensembled to predict 14 pathologies. In single-view cases (Stages 2 and 3), DenseNet121 handles AP-only, and ResNet50 handles LV-only, each trained on 80% of the respective subset, with 10% for validation and 10% for testing. Training employs the Adam optimizer, binary cross-entropy loss, a ReduceLROnPlateau scheduler, and early stopping, initialized with ImageNet pre-trained weights. For explainability and pathology localization, saliency maps are generated from trained multi-label classifiers using Grad-CAM. These maps are then filtered with contour area thresholds, highlighting regions most influential to each disease prediction and resulting in accurate bounding boxes over pathologies in both AP and LV views [30, 31].

2. Radiology Report Generation (RRG)

The RRG process is mediated by an intelligent agent which answers user queries on a real-time basis. When query is received, the agent calls the vision encoder to classify and localize findings in the given input image pair (AP and/or LV). These results are concatenated with the query, embedded using mx-bai-embed-large or CLIP (depending on the method: CAF, CLAF, or DGF), and employed to retrieve *top-k* relevant documents from vector database by means of the cosine-similarity. The agent then synthesizes a contextual prompt by integrating identified documents, classification results and a predefined template, which is zero-shot input into the LLM to generate a radiology report with a structure that follows clinical standards. Such a strategy allows factual consistency and mitigates hallucinations, providing detailed, explainable reports confirmed on datasets, such as MIMIC-CXR and IU-Xray.

D. Evaluations

The proposed framework for report generation is rigorously evaluated using a set of quantitative and clinical metrics to ensure robustness across varying data characteristics. For multi-label classification, the vision encoder is evaluated using the area under the receiver operating characteristic curve (AUC-ROC). **Quantitative Metrics Evaluation:** For MIMIC-CXR and IU-Xray datasets, the lexical (METEOR and ROUGE-L) and semantic performance (BERTScore) metrics are measured and compared using reference text against their AI generated reports. These metrics are excluded from ChestXray8 due to the absence of reference text. Additionally, the metrics of CheXpert Labeler automatic rule-based tool [20, 32] and Clinical Label Indicator (CLI) are used to extract the pathological labels from the AI generated reports for further validation on MIMIC-CXR in terms of precision (P), recall (R), and F1-score (F1). This evaluation isn't compatible with the IU-Xray and ChestXray8 datasets because they use different annotation schemas and feature distinct label sets. **Expert Clinical Validation:** For all datasets (MIMIC-CXR, IU-Xray, and ChestXray8), a panel of three expert radiologists further scores reports on scale of 0 (worst) to 10 (best) for clinical relevance and accurate context where a weighted average score is computed and reported. This method ensures a balanced assessment providing a gold

standard for clinical trustworthiness and supporting the framework's practical utility in real-world scenarios [33].

IV. EXPERIMENTAL RESULTS

A. Multi-vision Encoder Results

The performance of the image-view and pathology classification is evaluated for frontal, lateral, and combined views. For classifying frontal (AP) and lateral (LV) views, the DenseNet121 model performed exceptionally well. It achieved 100% training accuracy, 99.67% validation accuracy, and 99.0% test accuracy. This demonstrates the model's strong ability to differentiate between AP and LV views, a critical function for automated CAD systems and subsequent tasks. The multi-label classification performance is evaluated across three scenarios: AP, LV, and combined (AP and LV) using the ensemble strategy. For the AP scenario, the DenseNet121 model achieves an average AUC of 81.5%. In the LV scenario, the ResNet50 model shows promising results with an average AUC of 81.4%. The ensemble approach of fusing DenseNet121 and ResNet50 outperforms the individual models and other STOA studies with an average AUC of 83.1%, leveraging complementary features from both views and achieving the highest AUC scores across most pathologies, as depicted in Fig 3. ensemble model and Table 1 illustrates all three trained classifiers.

B. Multimodal Report Generation results

The multimodal RAG strategies show a significant improvement on the structured radiology report generation compared to the traditional methods. The effectiveness of the proposed methods is supported by quantitative metrics and qualitative approaches, along with expert-level clinical validation. Baseline RAG methods, RAG-Fusion and Compression established reference performance levels. Among the tested LLMs (LLaMA 3.2, Mistral-NEMO, LLaVA, and Gemma-2), LLaVA consistently delivered the highest baseline

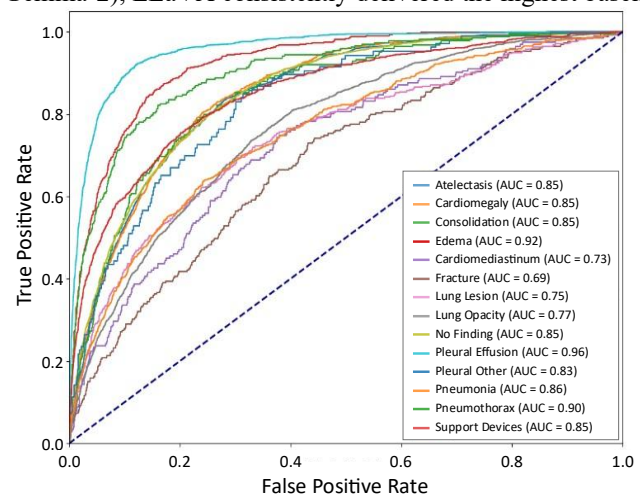


Fig 3. The area under the ROC curve (AUC) of the multi-label classification configurations for the 14 CXR pathologies of combined AP and LV views. The ensemble model demonstrates superior performance.

TABLE 1

COMPARISON OF AVERAGE AUC (%) FOR THE PROPOSED MULTI-LABEL CLASSIFICATION FRAMEWORK AGAINST RECENT STUDIES ACROSS MULTIPLE OF CXR PATHOLOGIES. THESE EVALUATION METRICS ARE RECORDED USING THE ENTIRE TESTING SET OF CXR IMAGES.

Respiratory Disease	Kalan et al. [12]	Haque et al. [11]	Li et al. [34]	AP View	LV View	AP & LV
Atelectasis	83.7	80.8	63.0	80.5	84.7	85.4
Cardiomegaly	82.8	81.6	63.0	80.8	83.9	85.9
Consolidation	84.4	82.1	78.0	81.3	82.5	85.6
Edema	90.4	88.9	78.0	88.7	91.7	92.7
Cardiomediastinum	75.7	73.9	75.0	71.7	73.9	73.7
Fracture	71.8	66.7	77.0	71.8	62.2	69.3
Lung Lesion	77.2	73.8	78.0	78.1	73.9	75.1
Lung Opacity	78.2	74.9	78.0	75.6	75.7	77.4
No Finding	86.8	85.3	70.0	85.3	84.4	85.9
Pleural Effusion	93.3	91.9	82.0	90.1	95.7	96.3
Pleural Other	84.8	80.6	82.0	84.1	81.7	83.4
Pneumonia	74.8	71.4	77.0	73.9	72.5	76.3
Pneumothorax	90.3	85.6	78.0	87.7	91.3	91.1
Support Devices	92.7	89.8	69.0	91.6	86.7	85.6

scores. On AP views, RAG-Fusion with LLaVA achieved METEOR 27.7%, ROUGE-L 27.7%, and BERTScore F1-score 84.1%. Compression, enhanced retrieval alignment, yielding METEOR 33.9%, ROUGE-L 32.7%, and BERTScore F1-score 85.2% on lateral views. While effective, these baselines fell short in handling multi-label dependencies across imaging views. **Table 2** illustrates the significant improvements delivered by the proposed methods. CAF, concatenating text and generated image-captioning, achieved a METEOR score of 35.8%, ROUGE-L of 35.2%, and BERTScore F1 of 85.4% on combined views. By precisely aligning the modalities using CLIP-based text-image embeddings, CLAF achieved a higher semantic fidelity, with Gemma-2 achieving METEOR 34.3%, ROUGE-L 32.2%, and BERTScore F1 86.1%. Reinforced by guided retrieval using additional docs, DGF factual grounding further improved METEOR 36.7%, ROUGE-L 32.1%, and BERTScore F1 84.7% augmented by LLaVA model. These results affirm that multimodal alignment enhances lexical precision and contextual continuity and exceeds baselines in all perspectives. CLI metric is further analyzed through label extraction from generated reports. Baseline approaches performed quite modestly, RAG-Fusion with precision 32.8%, F1 33.6%, and recall 34.3%; Compression with 36.5%, 37.4%, 38.2% respectively. In **Table 3**. CLI metrics for the proposed methods achieved outstanding results on the MIMIC dataset, especially in the F1-score values and recall. CAF matched precision of 40.8%, F1-score of 44.8% and recall of 49.7%.

TABLE 2

EVALUATION OF THE TEXT REPORT GENERATION (%) ACROSS VARIOUS RAG RETRIEVAL APPROACHES AND MULTI-LABELLED CLASSIFICATION TASKS. THE TESTING SET OF THE MIMIC-CXR DATASET IS USED FOR THIS EVALUATION.

VLLM	RAG Approach	Frontal Views (AP)					Lateral Views (AV)					Combined Views (AP & LV)				
		Meteor	Rouge-L	BertScore			Meteor	Rouge-L	BertScore			Meteor	Rouge-L	BertScore		
				P	F1	R			P	F1	R			P	F1	R
LLaMA 3.2	RAG Fusion	28.6	25.7	82.5	83.4	84.4	30.4	27.7	83.1	83.8	84.5	29.3	26.8	83.0	83.7	84.4
Mistral-NEMO		24.9	26.3	84.0	84.0	84.0	26.9	28.3	83.8	84.1	84.6	26.6	28.8	84.2	84.3	84.5
LLaVA		27.7	27.5	84.4	84.1	83.8	29.3	29.5	84.8	84.3	83.9	28.9	29.1	84.5	84.2	83.9
Gemma-2		26.3	25.0	82.8	84.0	85.2	24.7	23.0	81.0	82.7	84.5	25.6	24.0	81.3	82.9	84.6
LLaMA 3.2	Compression	28.0	27.4	84.4	84.0	83.6	30.7	29.1	85.1	84.6	84.1	30.8	29.3	85.0	84.5	84.1
Mistral-NEMO		25.7	27.3	84.5	84.4	84.4	30.2	31.7	85.2	84.1	85.9	29.0	30.9	85.0	84.9	84.9
LLaVA		29.1	28.2	84.3	84.2	84.2	33.9	32.7	85.2	85.2	85.2	34.2	32.5	85.1	85.1	85.2
Gemma-2		28.5	27.6	85.4	85.2	85.1	33.3	31.1	85.8	85.9	86.1	32.1	30.0	85.5	85.6	85.8
LLaMA 3.2	CAF	31.3	31.1	84.5	84.4	84.4	35.2	35.7	85.7	85.6	85.5	34.3	33.8	85.2	85.3	85.4
Mistral-NEMO		26.8	30.7	84.1	84.2	84.4	30.3	35.0	85.0	85.1	85.2	30.6	34.4	84.7	85.0	85.2
LLaVA		32.5	31.0	84.2	84.4	84.6	36.0	36.7	85.8	85.6	85.4	35.8	35.2	85.4	85.4	85.4
Gemma-2		30.8	31.1	85.6	85.6	85.6	35.3	35.0	86.5	86.5	86.5	30.1	34.0	84.7	85.0	85.2
LLaMA 3.2	CLAF	29.5	27.8	83.6	83.8	84.0	33.5	31.0	84.7	85.0	85.4	32.7	30.1	84.5	84.7	85.0
Mistral-NEMO		27.4	29.5	84.1	84.3	84.5	30.8	33.9	84.9	85.1	85.2	30.4	32.8	84.7	84.9	85.1
LLaVA		31.7	28.0	83.5	83.9	84.4	36.1	33.7	84.9	85.0	85.1	35.2	31.1	84.2	84.6	85.0
Gemma-2		30.5	29.5	85.0	85.3	85.7	34.7	32.9	86.0	86.2	86.4	34.3	32.2	85.8	86.1	86.4
LLaMA 3.2	DGF	32.4	30.4	83.9	84.2	84.5	34.7	31.8	84.4	84.8	85.2	34.7	31.6	84.3	84.8	85.4
Mistral-NEMO		28.1	30.5	83.8	84.2	84.5	31.4	33.7	84.4	84.8	84.2	30.5	32.6	84.3	84.7	85.2
LLaVA		34.4	30.2	83.6	84.1	84.7	37.0	33.8	84.6	84.9	85.4	36.7	32.1	84.2	84.7	85.3
Gemma-2		32.6	31.4	85.2	85.6	85.9	35.5	33.6	85.8	86.2	86.6	35.1	32.9	86.7	86.1	86.5

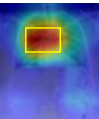
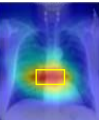
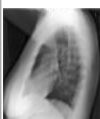
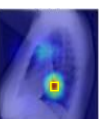
Input Image	Saliency Map	Ground-Truth Clinical Text Report	AI-based Structured Radiological Report	Experts' Validation (3 Radiologists)
		INDICATION: Chest pain to left shoulder, evaluate for pneumonia or effusion. COMPARISON: None. FINDINGS: There is no pleural effusion, pneumothorax or focal airspace consolidation. Heart size is normal. There is no evidence of pulmonary edema. The aorta is unfolded. IMPRESSION: No acute cardiopulmonary process.	EXAMINATION: CHEST (PA AND LAT) INDICATION: Routine follow-up of cardiac pacemaker TECHNIQUE: Chest PA and lateral views, portable technique COMPARISON: Previous studies from one year ago FINDINGS: The lungs are fully expanded and clear. Normal cardiomeastinal silhouettes with top-normal heart size unchanged compared to prior study. No pulmonary vascular congestion, edema, or pleural effusion. Transvenous right atrial and right ventricular pacemaker leads in stable positions, unchanged from previous studies. Degenerative pleural calcification noted at both apices, unchanged. IMPRESSION: No acute cardiopulmonary process. Stable position of cardiac pacemaker lead.	Evaluation: Meteor 43%, BertScore_F1-score 85%, Rouge-L 20% Expert Radiologist Evaluation Score: 10/10
★ Frontal (AP) ● No Finding ○ No Finding: 80%				
		HISTORY: Acute onset chest pain x. TECHNIQUE: Frontal and lateral views of the chest. COMPARISON: — FINDINGS: There is minimal left base atelectasis. No focal consolidation, pleural effusion, or evidence of pneumothorax is seen. The cardiac and mediastinal silhouettes are stable. No overt pulmonary edema is seen. Surgical hardware is seen in the cervical spine. No displaced fracture seen. IMPRESSION: Minimal left base atelectasis. Otherwise, acute cardiopulmonary process.	EXAMINATION: CHEST (PA AND LAT) INDICATION: Routine follow-up TECHNIQUE: Chest PA and lateral views COMPARISON: Studies from prior year available for comparison FINDINGS: Stable cardiac and mediastinal silhouettes. No pneumothorax or large pleural effusion identified. Patchy left base retrocardiac opacity, likely atelectasis, unchanged from previous studies. Subtle elevation of the right hemidiaphragm noted but clinically insignificant IMPRESSION: No acute cardiopulmonary abnormalities detected.	
★ Combine (AP & LV) ● Atelectasis ○ Atelectasis: 86%				
		INDICATION: Multiple myeloma, splinting from pain. COMPARISON: None. PA AND LATERAL VIEWS OF THE CHEST: Left-sided Port-A-Cath device is noted with tip terminating in the mid SVC. The heart size is normal. The mediastinal and hilar contours are unremarkable except for aortic knob calcifications. Pulmonary vascularity is normal. Linear opacities within both lung bases, more pronounced on the left, are compatible with subsegmental atelectasis. No pleural effusion or pneumothorax is present. Expansile lesion of the fifth anterior rib on the right may be related to underlying myeloma. IMPRESSION: Linear opacities at both lung bases, likely reflective of subsegmental atelectasis. Expansile heterogeneous lytic lesion in the right anterior fifth rib is likely related to known history of myeloma.	EXAMINATION: CHEST (PA AND LAT) INDICATION: Routine follow-up, low lung volumes TECHNIQUE: Chest PA and lateral views COMPARISON: Comparison made with prior studies from one year ago. FINDINGS: Low lung volumes bilaterally, more pronounced on the left. Streaky opacity in the retrocardiac region of the left base, likely due to atelectasis/scarring but infectious process not entirely excluded. Trace left pleural effusion suspected due to slight blunting of the left costophrenic angle and overlying soft tissue. Slight elevation of the right hemidiaphragm. No pneumothorax or displaced fracture seen. IMPRESSION: Low lung volumes with stable retrocardiac opacity likely representing atelectasis/scarring. Trace left pleural effusion suspected, but not significant. No acute cardiopulmonary process identified that would warrant urgent intervention. Dedicated PA and lateral views of the chest could help further evaluate the retrocardiac opacity if clinically indicated.	Evaluation: Meteor 35%, BertScore_F1-score 87%, Rouge-L 25% Expert Radiologist Evaluation Score: 10/10
★ Lateral (LV) ● Atelectasis ● Lung Opacity ○ Atelectasis: 55% ○ Lung Opacity: 66% ○ Pleural Effusion: 50%				
				Evaluation: Meteor 27%, BertScore_F1-score 83%, Rouge-L 18% Expert Radiologists' Evaluation Score: 08/10

Fig 4. Qualitative evaluation of the proposed CAD framework showing input images, saliency maps localization, ground-truth clinical reports, AI-generated structured reports, and expert radiologist validation. Red-highlighted red text indicates incorrect AI predictions removed during expert review [★ Image view Classification, ● Ground truth label, ○ Predicted label].

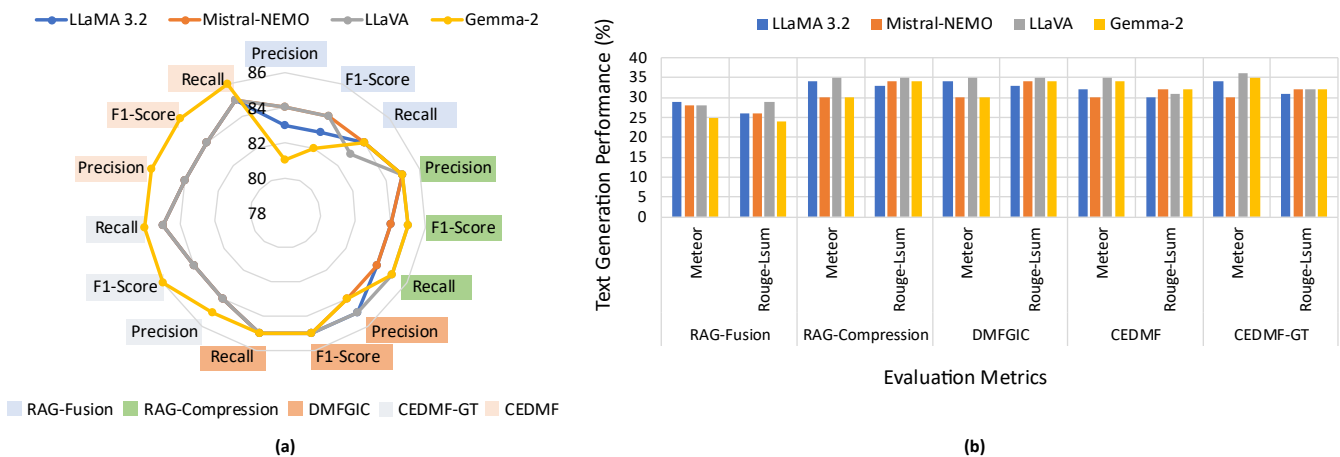


Fig 5. Comparative evaluation results of RRG using LLMs based on the baseline and proposed RAG approaches. (a) BERTScore Comparison, (b) METEOR and ROUGE-L Metrics.

DGF had F1-score of 43.7% and highest recall of 59.5%. This means higher extraction of clinically significant labels, thus increasing diagnostic reliability. Fig 4. shows a few samples of generated structured reports, ground-truth reports, saliency maps, and expert radiologist assessments with an emphasis on clinical relevance rather than automated metrics. A panel of

three experienced radiologists evaluated the test text reports from MIMIC-CXR, IU X-ray, and ChestX-ray8 datasets. The reports were rated on a 0 (bad) to 10 (excellent) scale, the weighted aggregation across datasets resulted in a weighted average expert score of 7.15 out of 10.0. This score indeed indicates that the proposed CAD system produces high-quality

TABLE 3
COMPARATIVE EVALUATION RESULTS OF RADIOLOGICAL REPORT GENERATION (RRG) AGAINST EXISTING STATE-OF-THE-ART RESEARCH WORKS.

	Dataset	Method of RRG	Structured Report	Evaluation metrics (%)							
				Meteor	Rouge-L	BertScore			Clinical Label Indicator (CLI)		
						P	F1	R	P	F1	R
Li et al. (2023), [34]	MIMIC-CXR	DCL	×	15.0	28.4	-	-	-	47	37	35
Liu et al. (2024), [35]		MiniGPT-4 (FT)	×	12.2	22.0	-	-	-	33.5	28.4	24.6
Liu et al. (2019), [36]		domain-aware automatic RRG	×	30.7	-	-	-	-	41	-	36
Wang et al. (2024), [16]		R2GenCSR-Llama2	×	16.7	29.1	-	-	-	-	-	-
Xia et al. (2024), [17]		MMed-RAG	×	20.4	12.3	-	-	-	-	-	-
Chen et al. (2020), [37]		R2Gen	×	14.2	27.7	-	-	-	33.3	27.6	27.3
Chen et al. (2022), [32]		R2GenCMN	×	14.2	27.8	-	-	-	33.4	27.8	27.5
Liu et al. (2021), [38]		PPKED	×	14.9	28.4	-	-	-	-	-	-
Yu. et al (2025), [39]		AHP-Full	×	15.4	28.5	-	-	-	-	-	-
Yan, B. et al (2022), [40]		ClinicalBERT	×	14.4	27.5	-	-	-	39.7	41.5	43.5
Wang. et al (2023), [41]		METrans	×	15.2	29.1	-	-	-	36.4	31.1	30.9
Thawkar et al (2023), [42]		XrayGPT	×	0.07	11.1	-	-	-	-	32.6	-
Proposed CAD System (2025)		LLaVA-based CAF	✓	35.8	35.2	85.4	85.4	85.4	40.8	44.8	49.7
		LLaVA-based CLAF	✓	35.2	31.2	84.2	84.6	85.0	34.2	39.0	45.2
		LLaVA-based DGF	✓	36.7	32.1	84.2	84.7	85.3	34.5	43.7	59.5
Li et al. (2023), [34]	IU Xray	DCL	The IU Xray dataset does not contain ground truth (GT) structured text reports, but only includes the FINDING section	19.3	38.3	-	-	-	-	-	-
Liu et al. (2024), [35]		MiniGPT-4 (FT)		20.8	39.0	-	-	-	-	-	-
Liu et al. (2019) [36]		domain-aware automatic RRG		35.9	-	-	-	-	-	-	-
Wang et al. (2024), [16]		R2GenCSR-Llama2		21.5	40.1	-	-	-	-	-	-
Xai et al. (2024), [17]		MMed-RAG		32.4	25.5	-	-	-	-	-	-
Gao et al. (2024), [43]		pathology-aware regional prompts		-	-	-	-	-	-	-	-
Chen et al. (2020), [37]		R2Gen		18.7	37.1	-	-	-	-	-	-
Chen et al. (2022), [32]		R2GenCMN		19.1	37.5	-	-	-	-	-	-
Liu et al. (2021), [38]		PPKED		19.0	37.6	-	-	-	-	-	-
Yu. et al (2025), [39]		AHP-Full		21.2	38.8	-	-	-	-	-	-
Proposed CAD System (2025)		LLaVA-based CAF		29.8	20.6	86.3	86.8	87.4	CLI measurements are not compatible with the IU-Xray dataset’s label schema		
		LLaVA-based CLAF		29.9	17.0	84.5	85.8	87.3			
		LLaVA-based DGF		30.6	16.9	83.8	85.6	87.5			

clinical reporting with results closely related to reference text interpretations with less potential of diagnostic misrepresentation. Overall, CAF, CLAF, and DGF have better benchmarks, label extraction, and verification of trained experts compared to the baselines.

V. DISCUSSION

This section presents the experimental results of the proposed multimodal retrieval-based CAD framework, focusing on respiratory disease prediction and radiology report generation. The evaluation encompasses both the diagnostic performance

TABLE 4

COMPARATIVE ANALYSIS OF REPORT GENERATION PERFORMANCE: MAIRA-2 MODEL VS. PROPOSED MULTIMODAL RAG APPROACHES ON MIMIC-CXR AND IU X-RAY DATASETS.

Dataset	Method of RRG	Evaluation metrics (%)				
		Meteor	Rouge-L	BertScore		
				P	F1	R
MIMIC-CXR	MAIRA2	18.9	19.1	85.7	83.4	80.8
	CAF	35.8	35.2	85.4	85.4	85.4
	CLAF	35.2	31.1	84.2	84.6	85.0
	DGF	36.7	32.1	84.2	84.7	85.3
IU X-ray	MAIRA2	25.1	19.0	80.1	83.5	87.3
	CAF	29.8	20.6	86.3	87.8	87.4
	CLAF	29.9	17.0	84.5	85.8	87.3
	DGF	30.6	16.9	83.8	85.6	87.5

TABLE 5

TEXT GENERATION EVALUATION PERFORMANCE (%) FOR THE FIRST ABLATION STUDY USING THE IU X-RAY DATASET

VLLM	RAG Method	Meteor	Rouge-L	BertScore		
				P	F1	R
LLaMA 3.2	RAG Fusion	15.6	08.8	77.2	80.0	83.2
Mistral-NEMO		10.3	10.2	83.2	83.2	83.2
LLaVA		15.6	11.0	82.3	82.5	82.9
Gemma-2		12.7	06.6	76.3	79.1	82.2
LLaMA 3.2	Compression	23.9	17.5	84.8	85.6	86.5
Mistral-NEMO		24.1	17.8	83.9	85.3	86.9
LLaVA		27.7	15.7	84.1	85.4	86.7
Gemma-2		16.8	13.7	83.6	84.0	84.6
LLaMA 3.2	CAF	26.5	22.1	84.9	86.0	87.1
Mistral-NEMO		22.1	21.9	84.1	86.4	85.8
LLaVA		29.8	20.6	86.3	87.8	87.4
Gemma-2		23.8	18.2	85.5	85.9	86.4
LLaMA 3.2	CLAF	24.2	17.2	83.5	84.9	86.4
Mistral-NEMO		22.8	19.0	83.3	84.9	86.6
LLaVA		29.9	17.0	84.5	85.8	87.3
Gemma-2		22.6	15.2	83.9	84.8	85.9
LLaMA 3.2	DGF	26.9	18.5	83.4	85.1	86.8
Mistral-NEMO		25.7	20.0	83.0	84.9	86.9
LLaVA		30.6	16.9	83.8	85.6	87.5
Gemma-2		25.6	16.9	83.8	85.1	86.5

of the multi-vision encoder in image classification and the effectiveness of the proposed RAG methods in generating clinically accurate and semantically coherent radiology reports, with comparisons against baseline approaches and ablation studies across multiple datasets.

A. Multi-vision Encoder

The proposed multi-vision encoder has outperformed at the classification task compared to other state-of-the-art methods, as summarized in **Table 1**. In comparison with previous studies, the framework performed better against most of the categories of respiratory disease. Pleural effusion had an AUC of 96.3%, and pneumothorax 91.1%, which is significantly higher compared to what occurred in previous studies. Likewise, better results were shown by atelectasis 85.4%, cardiomegaly 85.9% and pneumonia 76.3%. The overall average AUC of 83.1% substantiates the model as a competitive improvement of recent

Input Image	Saliency Map	AI-based Structured Radiological Report	Experts' Validation (Radiologists)
		Generated Report: Atelectasis is present in both lungs with a predominance in the right lower lobe and left base. There is also evidence of pleural effusion, specifically a moderate-sized right pleural effusion. The cardiomeastinal silhouette is normal. Clips are noted in the upper abdomen. No pneumothorax or focal consolidation is evident. Expert Radiologist Evaluation Score: 10/10	
		Generated Report: Cardiomegaly with pulmonary vascular congestion present, accompanied by new patchy bibasilar opacities. Severe cardiomegaly seen on PA and lateral chest radiographs. No focal consolidation, pleural effusion or pneumothorax. Cardiac, hilar and mediastinal contours are normal. Small pleural effusions noted. As compared to prior chest radiograph, short-term followup radiographs after diuresis may be helpful to exclude other superimposed processes such as infection. Expert Radiologist Evaluation Score: 10/10	
		Generated Report: The chest radiographs show a small to moderate-sized left bilateral pleural effusion which is stable in size. There is moderate pulmonary edema, consistent with diffuse interstitial opacity, which is also stable in severity. A new vague opacity in the right upper lobe is likely the sequela of pulmonary edema, though attention on follow-up radiographs is recommended. The cardiac silhouette remains moderately enlarged and the pulmonary arteries are markedly enlarged, unchanged. There is no pneumothorax. In the frontal chest radiograph, a persistent small right pleural effusion is noted. Linear left bibasilar opacities denote mild atelectasis, and there is no pneumothorax or focal consolidation. Clip in the left upper quadrant of the upper abdomen is present. Expert Radiologist Evaluation Score: 08/10	

Fig 6. Ablation study 2: Outcomes of the proposed CAD system using ChestX-ray8 dataset. Red-highlighted text indicates incorrect AI predictions removed during expert review [● Ground truth label, ● Predicted label].

baselines. Such improved results indicate the strength of the encoder in identifying discriminative AP and LV views that lead to consistent multi-label predictions capable of reinforcing RAG-based report generation downstream.

B. Multimodal Report Generation Comparative Analysis

The experiment results point out excellence of the proposed multimodal retrieval strategies (CAF, CLAF, and DGF) over baseline RAG methods. As demonstrated in **Fig 5(a)**, LLaVA-based CAF returns the best results of precision, recall, and F1-score, all with 85.4%. Likewise, DGF also increases retrieval precision using Gemma-2 scoring an 86.7 F1-score, which is more contextually consistent. LLaVA achieves much better METEOR and ROUGE-L text generation in **Fig 5(b)** with up to 37.0% and 36.7% improvement, respectively, over other LLMs. **Table 3**, presents a wider comparative view that indicates both lexical metrics and clinical label indicator (CLI) metrics. In comparison, the suggested CAF and DGF strategies are almost twice those numbers, with METEOR scores of 35.2%-36.7% and ROUGE-L consistently exceeding 30%. These lexical advances affirm superior structural and semantic correspondence with the ground truth reports. Notably, the CLI analysis on the MIMIC-CXR dataset shows that improvements in lexical performance are linked to clinically significant

TABLE 6

IMPACT OF SEMANTIC AND CLIP EMBEDDING DIMENSION ALIGNMENT ON RADIOLOGY REPORT GENERATION PERFORMANCE

(1) Semantic and CLIP Embeddings@1024-D					(2) Semantic and CLIP Embeddings@512-D				
Lexical Metric									
Meteor	Rouge-L	BertScore			Meteor	Rouge-L	BertScore		
		P	F1	R			P	F1	R
31.7	28.0	83.5	83.9	84.4	28.2	22.5	81.6	83.3	82.4
Clinical Label Indicator (CLI)									
P	F1	R			P	F1	R		
34.2	39.0	45.2			24.9	28.8	34.2		

TABLE 7

RETRIEVAL AND GENERATION COST TIME ACROSS VARIOUS RAG APPROACHES WITH DIFFERENT TOP-K SETTINGS. THE RESULTS ARE RECORDED AS AN AVERAGE OVER ALL TESTING SETS. (*) MEANS END-TO-END REPORT GENERATION OF THE PROPOSED CAD SYSTEM

RAG Approach	Document Retrieval Cost Time (Seconds)			RRG Cost time (Seconds)*		
	Top-k = 3	Top-k = 5	Top-k = 10	Top-k = 3	Top-k = 5	Top-k = 10
RAG-Fusion	5.95	6.04	10.84	41.87	61.48	122.75
Compression	10.54	15.56	26.01	33.38	34.73	52.76
CAF	0.02	0.01	0.05	16.95	21.39	31.85
CLAF	0.01	0.02	0.05	17.26	22.54	32.44
DGF	0.01	0.02	0.05	17.44	23.91	35.45

Datasets	Ground-Truth Clinical Text Report	AI-based Structured Radiological Report	Experts' Validation (Radiologists)
MIMIC CXR	HISTORY: ____-year-old male with left upper chest tenderness. STUDY: PA and lateral chest radiograph. COMPARISON: ____. FINDINGS: The cardiomeastinal and hilar contours are normal. The lungs are clear. There is no pleural effusion or pneumothorax. The visualized osseous structures show no evidence of injury. IMPRESSION: No acute cardiopulmonary process; no bony injury.	EXAMINATION: CHEST (PA AND LAT) INDICATION: Routine follow-up post-sternotomy TECHNIQUE: Chest PA and lateral without contrast COMPARISON: CT chest from ____ FINDINGS: Sternotomy wires intact. Interval improvement in pulmonary edema, predominantly affecting upper lobes with underlying fibrosis. Residual bilateral upper lobe and peripheral heterogeneous opacities with minimal interval change. No new focal consolidations to suggest infectious process. Normal heart size and mediastinal contour; unchanged hila. Emphysema, pleural calcifications, and diaphragmatic calcifications noted. No bony abnormalities IMPRESSION: No acute cardiopulmonary process or significant interval change. Interval improvement in pulmonary edema noted. Residual bilateral upper lobe opacities likely represent chronic interstitial lung disease. Normal heart size and mediastinal contour maintained.	Evaluation: Meteor 34%, BertScore_F1-score 85%, Rouge-L 18% Expert Radiologist Evaluation Score: 5/10
IU Xray	FINDINGS: No change lung XXXX. XXXX opacities are present in the right lower lobe. No focal infiltrates. Heart and mediastinum are unremarkable. Aorta normal. IMPRESSION: Findings of COPD with right lung base focal atelectasis. No evidence for failure or pneumonia.	FINDINGS: The heart size is top-normal. Mediastinum is normal. There is mild vascular congestion but no overt edema. Mild improvement in the lung bases variation is noted, but there is a persistent focal opacity in the right lower lung which might be concerning for an infectious process. Attention to this area on subsequent studies is recommended.	Evaluation: Meteor 20%, BertScore_F1-score 86%, Rouge-L 18% Expert Radiologist Evaluation Score: 4/10
ChestX-ray8	NO GROUND TRUTH	FINDINGS: There is a focal opacity in the right lower lobe that may represent a small mass or nodule. Otherwise, the lungs appear clear. The cardiomeastinal silhouette is normal. There is no evidence of pleural effusion or pneumothorax. Clips project over the left lung, potentially within the breast. The imaged upper abdomen is unremarkable. Chronic deformity of the posterior left sixth and seventh ribs are noted.	Expert Radiologists' Evaluation Score: 3/10

Fig 7. Examples of challenging cases with the lowest performance in report generation. Evaluation scores of Lexical metrics and expert validation (red font) are reported for each case from the MIMIC-CXR, IU X-ray, and ChestX-ray8 datasets.

improvements as well. CAF achieves a precision of 40.8%, an F1-score of 44.8%, and a recall of 49.7%, while DGF provides the best recall of 59.5% with an F1-score of 43.7%. These improvements indicate more accurate use of clinical words and labels, a critical aspect for trustworthy radiological reporting. **Table 4.** Additionally, the performance of the MAIRA2 model against our proposed approaches shows that while MAIRA2 establishes a competitive baseline, CAF and DGF substantially outperform it across both lexical and semantic metrics. Furthermore, expert validation scores were calculated for the generated reports, achieving an average score of 7.46/10 on the MIMIC-CXR dataset. In addition, empirical analysis was conducted as summarized in **Table 6** to investigate the impact of zero-padding for the CLIP feature embeddings to align the semantic embeddings. The results of both embeddings at 1024-D confirm that zero-padding of the CLIP embeddings to match the 1024-D of semantic space does not negatively impact the feature alignment or quality of the report generation. However, reducing the semantic embedding dimension from 1024-D to 512-D by pairwise averaging strategy to mimic the CLIP space

notably decreases the performance across both lexical and clinical evaluation metrics. These findings indicate that the full 1024-D semantic embedding retains critical contextual information necessary for accurate and clinically coherent radiology report generation.

C. Ablation Study 1

On the IU X-ray dataset, where only the "FINDING" sections are available, the proposed retrieval methods were benchmarked against baselines. As shown in **Table 5**, advanced multimodal approaches consistently outperform traditional RAG methods. CAF achieved the best results, with a METEOR score of 29.8%, ROUGE-L of 20.6%, and an F1-score of 87.8%, reflecting strong semantic alignment with ground-truth reports. CLAF and DGF also demonstrated competitive results, highlighting their robustness in preserving clinical relevance despite limited structured labels. Additionally, expert validation scores were calculated for the generated reports achieving an average score of 5.9/10 on the IU X-ray dataset, further

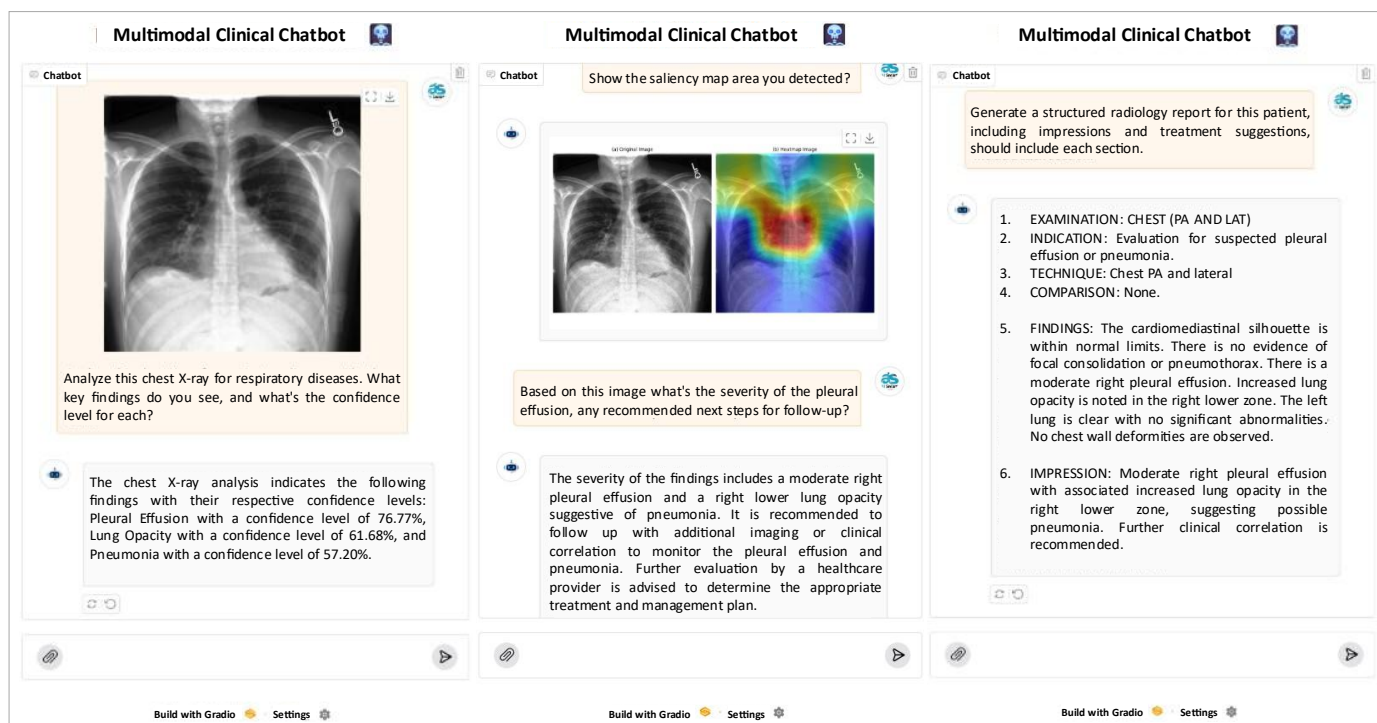


Fig 8. Chatbot-in-Action: analyzing the chest X-ray images for key findings, saliency mapping, severity assessment, and structured reporting

confirming the clinical reliability of our multimodal retrieval approaches.

D. Ablation Study 2

The ChestX-ray8 dataset, which lacks ground-truth text, was used to test the adaptability of the CAD system scenarios without annotated reports. In this case, lexical metrics cannot be computed; instead, evaluation relied on expert radiologist validation of generated reports, as illustrated in Fig 6. The generated structured reports received high clinical scores, with two cases rated 10/10 and one case rated 08/10. The expert validated testing reports achieving an average score of 7.4/10. These expert evaluations confirm that even in the absence of textual ground truth, the proposed system produces clinically useful outputs that align with diagnostic expectations.

E. Analysis of Retrieval and Generation Approaches

Computational efficiencies and stability of proposed retrieval methods are summarized in Table 7. Compared to the baseline RAG methods, our methods are significantly faster at document retrieval. Selection of the $top-k$ value is a critical factor in this process, as it influences machine stability as well as resource utilization. An analysis of the average scores across all approaches reveals remarkable consistency in performance for $top-k$ values of 3, 5, and 10. For instance, the average Meteors are 34.52, 34.56, and 34.96 for $top-k=3$, $top-k=5$, and $top-k=10$, respectively. This suggests that increasing the number of documents retrieved above five has a negligible effect of improving quality of generation in our case. So, higher $top-k$ values will mainly bring increased computational costs with no significant improvement. Thus, we used $top-k=5$ as the most suitable parameter in our scenarios. It balances document diversity and efficient processing time ideally. Among the RAG techniques, CAF reaches the fastest retrieval and generation times. At $top-k=5$, it takes only 0.01 seconds to finish document

retrieval and produces the final report in 21.39 seconds. These findings underscore the effectiveness of CAF in retrieving a high-volume of information in a time-optimal manner. Such configurations are therefore necessary to enable the proposed CAD system to achieve optimal prediction accuracy but also reduce the retrieval and generation time.

F. Limitations and Future Work

Although the framework has made significant progress, limitations remain. In particular, the relatively modest ROUGE-L and METEOR scores highlight the limitation of lexical metrics in fully capturing clinical accuracy and semantic equivalence, underscoring the need for domain-specific evaluation methods to better assess factual and diagnostic coherence. It can be vulnerable to missing ground-truth, as with datasets like ChestX-ray8 that lack paired clinical records; reliance on textual embeddings for data retrieval may exaggerate hallucinations or fail to properly ground contextual knowledge in such cases. Furthermore, as shown in Fig 7., the worst-case analysis captures the cases in which the framework sometimes misreports or overreports secondary findings, despite strong lexical metrics, which could prompt unnecessary clinical follow-ups if not checked. This underscores the need for a human-in-the-loop process, in which radiologists review AI-generated drafts before finalization in high-risk or low-confidence cases. An effective clinical workflow would involve automatically flagging cases with high label uncertainty or low confidence for specialist review, ensuring that the final diagnosis and responsibility remain with the clinician. To support this, we have developed a proof-of-concept to query and tweak outputs in real-time, as Fig 8. illustrates, where disease classification and saliency maps, severity assessments with recommendations, and complete reports become a single fluid exchange, simplifying clinician adjustments. Future work

will overcome these limitations using unsupervised methods to synthetically generate reports on text-constrained datasets, and by extending the chatbot to standalone X-ray deployment with enhanced personalization for routine clinical use.

VI. CONCLUSION

This study introduces a pioneering multimodal retrieval CAD framework that fuses a vision language model with advanced RAG techniques to advance respiratory disease prediction and structured radiology report generation. By pioneering the integration of multimodal retrieval with zero-shot structural reporting, it overcomes limitations of unimodal systems, delivering exceptional diagnostic accuracy, reduced hallucinations, and enhanced interpretability through novel strategies like CAF, CLAF, and DGF. The multi-vision encoder, adding DenseNet121 and ResNet50, excels in binary view and multi-label pathology classifications. Expert evaluations and superior metrics across datasets affirm their real-world viability, setting a new standard for explainable AI in medicine and paving the way for transformative clinical applications.

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